Student Record of Today's Lecture

Double Integrals in Polar Coordinates and Change of Order

1. Polar double integrals

Polar coordinate conversion

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad x^2 + y^2 = r^2.$$

Usually $r \geq 0$, while θ describes direction from the positive x-axis.

Why $dA = r\,dr\,d\theta$

A very small polar cell has radial thickness dr. At distance r from the origin, a small angular change $d\theta$ corresponds to arc length approximately

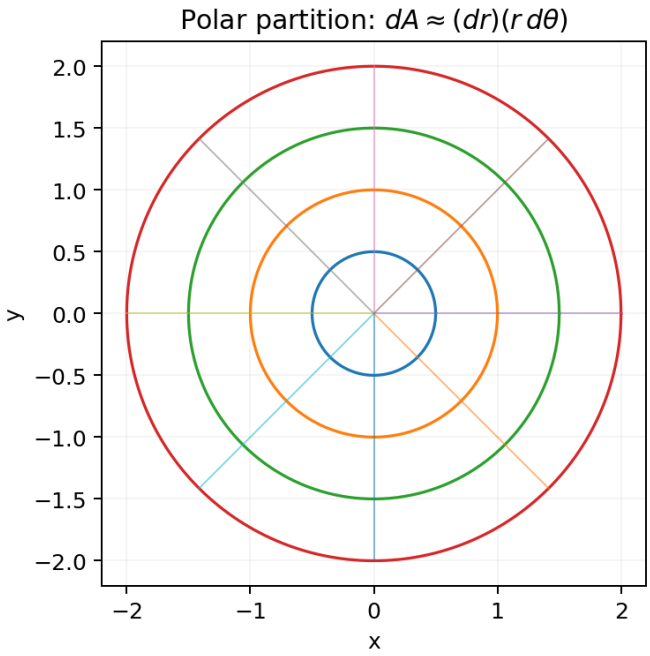
$$r\,d\theta.$$

Hence

$$dA \approx (dr)(r\,d\theta),$$

so

$$\boxed{dA = r\,dr\,d\theta}.$$



If

$$\alpha \leq \theta \leq \beta, \qquad r_1(\theta) \leq r \leq r_2(\theta),$$

then

$$\boxed{\iint_R f(x,y)\,dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta.}$$

Example 20.1 - Area of a disk

Use a double integral to find the area of

$$x^2 + y^2 \leq R^2.$$

Detailed solution

The disk is

$$0 \leq r \leq R, \qquad 0 \leq \theta \leq 2\pi.$$

Therefore

$$A = \iint_D 1\,dA = \int_0^{2\pi} \int_0^R r\,dr\,d\theta.$$

Thus

$$A = \int_0^{2\pi} \left[\frac{r^2}{2}\right]_0^R d\theta = \int_0^{2\pi} \frac{R^2}{2}\,d\theta = \boxed{\pi R^2}.$$

Example 20.2 - Radial integrand over a disk

Evaluate

$$\iint_D (x^2+y^2)\,dA, \qquad D : x^2+y^2 \le 4.$$

Detailed solution

Use

$$x^2+y^2 = r^2, \qquad dA = r\,dr\,d\theta,$$

with

$$0 \le r \le 2, \qquad 0 \le \theta \le 2\pi.$$

Hence

$$\iint_D (x^2+y^2)\,dA = \int_0^{2\pi} \int_0^2 r^3\,dr\,d\theta.$$

Now

$$\int_0^2 r^3\,dr = \left[\frac{r^4}{4}\right]_0^2 = 4.$$

Therefore

$$\boxed{\iint_D (x^2+y^2)\,dA = 8\pi}.$$

Example 20.3 - Quarter-annulus

Evaluate

$$\iint_R xy\,dA$$

over the first-quadrant quarter-annulus

$$1 \le x^2+y^2 \le 9.$$

Detailed solution

The region becomes

$$1 \le r \le 3, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

Also

$$xy = r^2 \sin\theta\cos\theta.$$

Thus

$$\iint_R xy\,dA = \int_0^{\pi/2} \int_1^3 r^3 \sin\theta\cos\theta\,dr\,d\theta.$$

Separate the factors:

$$= \left(\int_1^3 r^3\,dr\right)\left(\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta\right).$$

Now

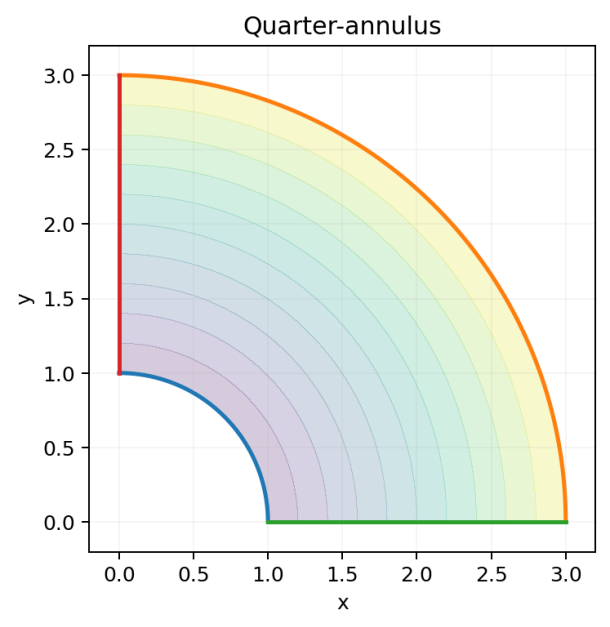
$$\int_1^3 r^3\,dr = \left[\frac{r^4}{4}\right]_1^3 = 20,$$

and

$$\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta = \frac{1}{2}.$$

Therefore

$$\boxed{\iint_R xy\,dA = 10}.$$



When polar coordinates are natural

Think of polar coordinates when the region or integrand contains circles, disks, annuli, sectors, radial symmetry, or repeated occurrences of $x^2 + y^2$. Always convert both the function and the area element.

2. Change of order of integration

Changing order means **describing the same region using the opposite slice direction**. It is not obtained by simply exchanging dx and dy.

Five-step method

1. Sketch the region from the original bounds.
2. Mark the boundary curves and intersection points.
3. Find the complete range of the new outer variable.
4. For a fixed value of that variable, identify where the new inner slice enters and leaves the region.
5. Split the region if the boundary formula changes.

Example 21.1 - Reversal without splitting

Change the order in

$$\int_0^1 \int_{x^2}^{x} f(x,y)\,dy\,dx.$$

Detailed solution

The original bounds describe

$$0 \le x \le 1, \qquad x^2 \le y \le x.$$

Thus the region lies between

$$y = x^2 \qquad \text{and} \qquad y = x.$$

Its total y-range is

$$0 \le y \le 1.$$

Solve both boundary equations for x in the first quadrant:

$$y = x \Longrightarrow x = y,$$

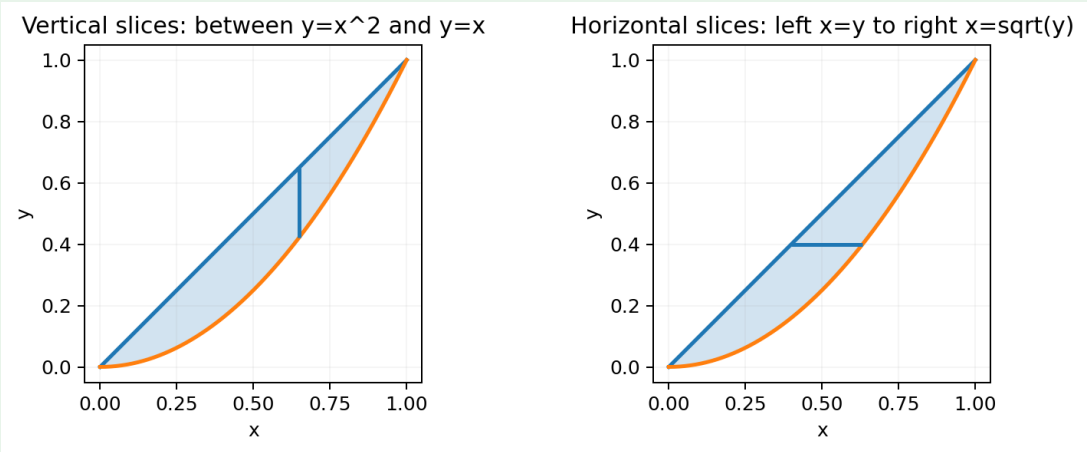
$$y = x^2 \Longrightarrow x = \sqrt{y}.$$

For fixed y, a horizontal slice therefore runs from

$$x = y \quad \text{to} \quad x = \sqrt{y}.$$

Hence

$$\boxed{\int_0^1 \int_{x^2}^{x} f(x,y)\,dy\,dx = \int_0^1 \int_{y}^{\sqrt{y}} f(x,y)\,dx\,dy.}$$



Example 21.2 - Evaluate after reversing

Evaluate

$$I = \int_0^1 \int_{x^2}^{x} (x + y)\, dy\, dx$$

by reversing the order first.

Detailed solution

From Example 21.1,

$$I = \int_0^1 \int_y^{\sqrt{y}} (x + y)\, dx\, dy.$$

The inner integral is

$$\int (x + y)\, dx = \frac{x^2}{2} + yx.$$

Therefore

$$\int_y^{\sqrt{y}} (x + y)\, dx = \frac{y}{2} + y^{3/2} - \frac{3y^2}{2}.$$

Hence

$$I = \int_0^1 \left(\frac{y}{2} + y^{3/2} - \frac{3y^2}{2} \right) dy.$$

Thus

$$I = \left[\frac{y^2}{4} + \frac{2}{5} y^{5/2} - \frac{1}{2} y^3 \right]_0^1 = \frac{1}{4} + \frac{2}{5} - \frac{1}{2} = \boxed{\frac{3}{20}}.$$

Change-of-order checklist

original bounds → sketch → new outer range → new inner entry/exit

If the entry or exit boundary changes, split the new outer-variable range.

Quiz focus

The announced quiz scope is **double integrals for finding areas and volumes, and change of order of integration**. Polar-coordinate integration was covered in today's lecture, but it is not listed among the announced quiz topics.